Diagonalisation of a Matrix

Class notes for B.Tech First Year (Semester I)
Engineering Mathematics - I (Matrices and Calculus)

What is inside

1. Words we need first (quick recall)

2. What is diagonalisation?

3. Why it works (the idea behind it)

4. Important notes

5. Working rule (5 steps)

6. Easy methods for eigenvalues and eigenvectors (no hard equations!)

7. Finding powers A^n using diagonalisation

8. Example 1: a 2 × 2 matrix (start here)

9. Example 2: a 3 × 3 matrix with a repeated eigenvalue

10. Example 3: a 3 × 3 matrix, find A^4

11. Summary, common mistakes and practice questions

1. Words we need first

Before we start, let us recall five small ideas. Each one is used again and again in this topic.

Idea 1: Diagonal matrix

A square matrix is **diagonal** if every number outside the main diagonal (top-left to bottom-right) is zero. Only the diagonal numbers can be non-zero. We write diag(a, b, c) for such a matrix.

$$\text{diag}(2, 5) = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \qquad \text{diag}(1, 2, 3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Idea 2: Eigenvalue and eigenvector

For a square matrix A, if a **non-zero** column X satisfies **AX = λX** for some number λ, then λ is an **eigenvalue** of A and X is the **eigenvector** for that λ. In simple words: when A multiplies X, the answer is just X stretched by the number λ. The direction of X does not change.

Idea 3: How to find eigenvalues

The eigenvalues are the roots of the **characteristic equation** $|A - \lambda I| = 0$. Section 6 gives an easy shortcut, so you do not need to expand big determinants.

Idea 4: How to find eigenvectors

For each eigenvalue λ, the eigenvector X solves $(A - \lambda I)X = 0$. Section 6 gives a direct method, so you do not need to solve equations by hand.

Idea 5: Inverse and linear independence

P^{-1} (inverse of P) satisfies $P^{-1}P = I$. It exists only when $|P| \neq 0$. We use $P^{-1} = \text{adj } P / |P|$. For a 2 × 2 matrix with entries a, b, c, d:

$$P = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{gives} \quad P^{-1} = (1 / |P|) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{where } |P| = ad - bc$$

Vectors are **linearly independent** if none of them can be made from the others by multiplying and adding. For two vectors this simply means: one is **not a multiple** of the other.

2. What is diagonalisation?

The main idea

A diagonal matrix is very easy to work with. For example, if D = diag(2, 5), then D^2 = diag(4, 25) and D^3 = diag(8, 125). We just raise each diagonal number to the power. No difficult multiplication!

A normal matrix is not like this. Finding A^8 by multiplying A eight times is slow and full of mistakes. **Diagonalisation** is a way to change a normal matrix A into a diagonal matrix D, so that hard work becomes easy work.

Definition

A square matrix A of order n is **diagonalisable** if we can find an **invertible** matrix P such that

P^{-1} A P = D, where D is a diagonal matrix.

We then say: "P **diagonalises** A" or "A is **transformed to diagonal form**".

Three names to remember

- **Similar matrices:** Two square matrices A and B of the same order are similar if there is an invertible matrix P with $B = P^{-1}AP$. We write $A \sim B$.
- **Similarity transformation:** the change Y = PX, which turns A into $P^{-1}AP$.
- **So** "A is diagonalisable" means "A is similar to a diagonal matrix".

Modal matrix and spectral matrix

The matrix P is called the **modal matrix** of A. Its columns are the eigenvectors of A.

The diagonal matrix D is called the **spectral matrix** of A. Its diagonal numbers are the eigenvalues of A.

Important: the eigenvalues in D must be in the **same order** as the eigenvectors in the columns of P.

3. Why it works (the idea behind it)

Students often ask: "Why do we put eigenvectors in P?" Let us see it slowly, with a 2 × 2 matrix. Let A have eigenvalues λ_1, λ_2 and eigenvectors X_1, X_2. That means $AX_1 = \lambda_1 X_1$ and $AX_2 = \lambda_2 X_2$.

Step 1: Build P from the eigenvectors

Write X_1 and X_2 side by side as the two columns of P: $P = [\, X_1 \; X_2 \,]$.

Step 2: Multiply A with P

A multiplies each column of P separately:

$$AP = [\, AX_1 \quad AX_2 \,] = [\, \lambda_1 X_1 \quad \lambda_2 X_2 \,]$$

Step 3: Write this as P times a diagonal matrix

When a diagonal matrix is on the **right** side of P, it multiplies the **columns** of P. So the first column gets multiplied by λ_1 and the second column by λ_2:

$$[\, X_1 \quad X_2 \,] \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = [\, \lambda_1 X_1 \quad \lambda_2 X_2 \,]$$

Therefore **AP = PD**, where $D = \text{diag}(\lambda_1, \lambda_2)$.

Step 4: Remove P from the right side

If P has an inverse, multiply both sides from the left by P^{-1}:

$$P^{-1}(AP) = P^{-1}(PD) = (P^{-1}P)D = I\,D = D$$

Result

P^{-1} A P = D. The same works for n × n matrices: AP = PD, with $D = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.

Where is the condition? In Step 4 we needed P^{-1} to exist. P has an inverse only if its columns are linearly independent. So we need **n linearly independent eigenvectors** for an n × n matrix. If we cannot find them, the matrix cannot be diagonalised.

4. Important notes

Main condition

A matrix A of order n can be diagonalised **only if** we can find **n linearly independent eigenvectors** of A. Then P is made from them and $P^{-1}AP = D$.

Notes

- **Note 1.** If the eigenvectors $X_1, \ldots, X_n$ are **not** linearly independent, then A is **not** diagonalisable.
- **Note 2.** If all the eigenvalues are **different**, the eigenvectors are always linearly independent, so A is always diagonalisable.
- **Note 3.** If an eigenvalue is **repeated**, A may or may not be diagonalisable. We must check whether we can still get enough independent eigenvectors. (Example 2 works. Practice question 2 fails.)
- **Note 4.** The change of A into diagonal form using $P^{-1}AP$, with P non-singular, is called a **similarity transformation**.
- **Note 5 (special case).** Let A be a **real symmetric** matrix with n different eigenvalues. Then the eigenvectors $X_1, \ldots, X_n$ are **pairwise orthogonal** (the dot product of any two is 0).

In that case divide each eigenvector by its length: $e_1 = X_1/\|X_1\|$, $e_2 = X_2/\|X_2\|$, ... and put $P = (e_1, e_2, \ldots, e_n)$. Then P is an **orthogonal matrix**, that is, $P^TP = PP^T = I$, so $\mathbf{P^{-1} = P^T}$. Hence $P^{-1}AP = D$ becomes $\mathbf{P^TAP = D}$. We do not need to find an inverse at all.

5. Working rule to diagonalise a matrix A

The 5 steps

Step 1: Find the eigenvalues of the square matrix A. (Use Shortcut A of Section 6.)

Step 2: Find the eigenvector for each eigenvalue. (Use Shortcut B or C of Section 6.) Write them as the columns of the modal matrix $P = [X_1, X_2, \ldots, X_n]$.

Step 3: Find P^{-1} using $P^{-1} = \text{adj } P / |P|$.

Step 4: Find the diagonal matrix $D = P^{-1}AP$. It should be $\text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.

Step 5: If asked, find A^n using $A^n = P\,D^n\,P^{-1}$.

Useful checking tricks

- **Sum check:** sum of eigenvalues = sum of diagonal numbers of A (the trace).
- **Product check:** product of eigenvalues = |A| (the determinant).
- **Eigenvector check:** put each X back in AX and see that you get λX.
- **Final check without inverse:** verify AP = PD. If this is true, then $P^{-1}AP = D$ is also true.
- **Order:** if you change the order of the columns in P, change the order of the numbers in D in the same way.

6. Easy methods for eigenvalues and eigenvectors

Many students find it hard to solve for eigenvalues and eigenvectors, especially for 3 × 3 matrices. Here are three shortcuts. With them there are no long equations to solve. Use them in every example.

Shortcut A: Eigenvalues

Formula

For a 2 × 2 matrix: $\lambda^2 - (S_1)\,\lambda + |A| = 0$, where S_1 = sum of the diagonal numbers.

For a 3 × 3 matrix: $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

S_1 = sum of the diagonal numbers (called the trace).

S_2 = sum of three small 2 × 2 determinants. For each diagonal number, cross out its row and its column, and take the determinant of what is left.

$S_3 = |A|$, the determinant of A.

How to solve the cubic equation (3 easy steps):

- **Find one root by trial.** Try $\lambda = 1, -1, 2, -2, 3, -3$ ... (numbers that divide the last number S_3). Put each in the equation until you get 0.
- **Divide** by that root using the small table below. This turns the cubic into a quadratic.
- **Solve the quadratic** by factorising. You now have all three eigenvalues.

Demo of the table. Equation $\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$. Try $\lambda = -3$: $(-27) + 9 + 63 - 45 = 0$. So $\lambda = -3$ is a root. Now divide:

Coefficients	1	1	−21	−45
Multiply by −3		−3	6	45
Add	1	−2	−15	0

How to fill it: bring down the first coefficient (1). Multiply it by −3 and write it under the next coefficient. Add. Repeat. The last answer is 0, which proves −3 is a root. The numbers 1, −2, −15 are the quadratic: $\lambda^2 - 2\lambda - 15 = (\lambda - 5)(\lambda + 3)$. So the roots are −3, 5, −3.

Two quick checks: sum of eigenvalues = S_1, and product of eigenvalues = S_3.

Shortcut B: Eigenvectors by cross multiplication

Idea. Each row of $(A - \lambda I)$ must "multiply" X to give 0. For a 3×3 matrix, take **any two rows** of $(A - \lambda I)$. Call them $R_1 = (a_1, a_2, a_3)$ and $R_2 = (b_1, b_2, b_3)$. Then the eigenvector $X = (x, y, z)$ is given directly by:

Cross-multiplication rule

$$\mathbf{x = a_2b_3 - a_3b_2}$$

$$\mathbf{y = a_3b_1 - a_1b_3}$$

$$\mathbf{z = a_1b_2 - a_2b_1}$$

Easy way to remember (cover-up): to get x, cover column 1 and take the 2×2 determinant of what is left. To get z, cover column 3 and do the same. To get y, cover column 2, take the determinant, and **change its sign** (that is why y is $a_3b_1 - a_1b_3$).

Why does it work? Put the answer in $R_1 \cdot X$: $a_1(a_2b_3 - a_3b_2) + a_2(a_3b_1 - a_1b_3) + a_3(a_1b_2 - a_2b_1)$. All six terms cancel in pairs and the answer is 0. The same happens with R_2. So X satisfies both equations.

Rules to follow:

- Choose two rows that are **not multiples** of each other, and with small numbers.
- You may divide a row by a common number first (for example $(2, -4, -6)$ becomes $(1, -2, -3)$).
- At the end you may divide the answer by a common number, or multiply it by -1. It is still an eigenvector.
- **Always check:** A X should be λ X.
- **For a 2×2 matrix it is even easier:** take one row (a, b) of $(A - \lambda I)$. Then $X = (b, -a)$.

Shortcut C: When the eigenvalue is repeated and only one equation is left

Sometimes all rows of $(A - \lambda I)$ are multiples of each other, and the cross multiplication gives $(0, 0, 0)$. Then we have only **one** equation $ax + by + cz = 0$ (with $a \neq 0$), and we need **two** eigenvectors. Use:

Rule for one equation

$X = (b, -a, 0)$ and **$X' = (c, 0, -a)$**

Check: put X in the equation: $a \cdot b + b \cdot (-a) + 0 = 0$. Put X': $a \cdot c + 0 + c \cdot (-a) = 0$. Both work, and they are not multiples of each other, so they are independent. (If $a = 0$, use any non-zero coefficient in the place of a.)

Example. $x + 2y - 3z = 0$ has $a = 1$, $b = 2$, $c = -3$. So $X = (2, -1, 0)$ and $X' = (-3, 0, -1)$, or $(3, 0, 1)$ after multiplying by -1.

7. Finding powers of a matrix: $A^n = P\,D^n\,P^{-1}$

We start from $D = P^{-1}AP$. Now square it:

$$D^2 \;=\; (P^{-1}AP)(P^{-1}AP) \;=\; P^{-1}A(PP^{-1})AP \;=\; P^{-1}A^2P \qquad \text{(because } PP^{-1} = I\text{)}$$

In the same way $D^3 = P^{-1}A^3P$, and in general $\mathbf{D^n = P^{-1}\,A^n\,P}$. To get A^n alone, multiply by P on the left and by P^{-1} on the right:

$$P\,D^n\,P^{-1} \;=\; (PP^{-1})\,A^n\,(PP^{-1}) \;=\; A^n$$

Result

$\mathbf{A^n = P\,D^n\,P^{-1}}$, where $D^n = \text{diag}(\lambda_1^n, \lambda_2^n, \ldots, \lambda_n^n)$.

Why is this good? We never multiply A by itself many times. We only raise the diagonal numbers to a power and then multiply three matrices: P, D^n and P^{-1}.

8. Example 1: a 2 × 2 matrix

Question

Diagonalise the matrix below and hence find A^3.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Step 1: Find the eigenvalues (Shortcut A)

For a 2 × 2 matrix: $\lambda^2 -$ (sum of diagonal) $\lambda + |A| = 0$.

Sum of diagonal = 4 + 3 = 7. $|A| = (4)(3) - (1)(2) = 10$.

So $\lambda^2 - 7\lambda + 10 = 0$, that is, $(\lambda - 2)(\lambda - 5) = 0$. **$\lambda_1 = 2$ and $\lambda_2 = 5$.**

Check: sum = 2 + 5 = 7 and product = 2 × 5 = 10. Both match. The two eigenvalues are different, so we will get two independent eigenvectors and A is diagonalisable.

Step 2: Find the eigenvectors (Shortcut B)

For a 2 × 2 matrix take any one row (a, b) of $(A - \lambda I)$. Then X = (b, −a).

For $\lambda = 2$. $A - 2I$ is:

$$A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

Take the first row (a, b) = (2, 1). Then X = (b, −a) = (1, −2). **$X_1 = [\ 1, -2\ ]^T$.**

Check: $AX_1 = [\ 4(1) + 1(-2), 2(1) + 3(-2)\ ]^T = [\ 2, -4\ ]^T = 2X_1$. Correct.

For $\lambda = 5$. $A - 5I$ is:

$$A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

Take the first row (a, b) = (−1, 1). Then X = (b, −a) = (1, 1). **$X_2 = [\ 1, 1\ ]^T$.**

Check: $AX_2 = [\ 4 + 1, 2 + 3\ ]^T = [\ 5, 5\ ]^T = 5X_2$. Correct.

Modal matrix: put X_1 and X_2 as columns.

$$P = [\ X_1 \quad X_2\ ] = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$$

Step 3: Find P^{-1}

$|P| = (1)(1) - (1)(-2) = 1 + 2 = 3$. This is not zero, so P^{-1} exists.

Use the 2×2 rule (swap the diagonal numbers, change the sign of the other two, divide by $|P|$):

$$P^{-1} = (1/3)\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

Step 4: Find $D = P^{-1}AP$

First find AP:

$$AP = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ -4 & 5 \end{bmatrix}$$

(For example, first row, first column: $4(1) + 1(-2) = 2$.) Now multiply by P^{-1}:

$$P^{-1}AP = (1/3)\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & 5 \\ -4 & 5 \end{bmatrix} = (1/3)\begin{bmatrix} 6 & 0 \\ 0 & 15 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

So **$D = \text{diag}(2, 5)$**. The diagonal numbers are the eigenvalues, in the same order as the columns of P. **A is diagonalised.**

Step 5: Find A^3

$D^3 = \text{diag}(2^3, 5^3) = \text{diag}(8, 125)$.

$A^3 = P\,D^3\,P^{-1}$. First $P\,D^3$ (multiply the first column of P by 8 and the second column by 125):

$$P\,D^3 = \begin{bmatrix} 8 & 125 \\ -16 & 125 \end{bmatrix}$$

Now multiply by P^{-1}:

$$A^3 = (1/3)\begin{bmatrix} 8 & 125 \\ -16 & 125 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = (1/3)\begin{bmatrix} 258 & 117 \\ 234 & 141 \end{bmatrix} = \begin{bmatrix} 86 & 39 \\ 78 & 47 \end{bmatrix}$$

Check by direct multiplication: $A^2 = [[18, 7], [14, 11]]$ and $A^3 = A^2 \cdot A = [[86, 39], [78, 47]]$. Same answer.

Answer

$A^3 = [[86, 39], [78, 47]]$. Direct multiplication was easy for power 3, but for A^{10} or A^{20} the formula PD^nP^{-1} is far quicker.

Class tip: ask students to swap the columns, $P = [X_2\ X_1]$, and see that D becomes $\text{diag}(5, 2)$.

9. Example 2: a 3 × 3 matrix with a repeated eigenvalue

Question

Determine the modal matrix P of A below. Verify that $P^{-1}AP$ is a diagonal matrix.

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

What is new here: one eigenvalue is repeated. We must check that we still get 3 linearly independent eigenvectors.

Step 1: Find the eigenvalues (Shortcut A)

S_1 = sum of diagonal = $-2 + 1 + 0 = -1$.

S_2 = sum of the three 2×2 determinants:

cross out row 1, column 1: $(1)(0) - (-6)(-2) = -12$

cross out row 2, column 2: $(-2)(0) - (-3)(-1) = -3$

cross out row 3, column 3: $(-2)(1) - (2)(2) = -6$

So $S_2 = -12 - 3 - 6 = -21$.

S_3 = $|A| = -2(0 - 12) - 2(0 - 6) + (-3)(-4 + 1) = 24 + 12 + 9 = 45$.

The equation $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ becomes:

$$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

Trial: try $\lambda = -3$: $-27 + 9 + 63 - 45 = 0$. So $\lambda = -3$ is a root. Divide:

Coefficients	1	1	−21	−45
Multiply by −3		−3	6	45
Add	1	−2	−15	0

The quadratic is $\lambda^2 - 2\lambda - 15 = (\lambda - 5)(\lambda + 3) = 0$.

Eigenvalues: $\lambda = 5$, $\lambda = -3$, $\lambda = -3$. (-3 is repeated twice.)

Check: sum $= 5 - 3 - 3 = -1 = S_1$. Product $= 5 \times (-3) \times (-3) = 45 = S_3$. Correct.

Step 2: Find the eigenvectors (Shortcuts B and C)

For λ = 5. $A - 5I$ is:

$$A - 5I = \begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix}$$

Choose two rows with small numbers. Row 2 divided by 2 is $(1, -2, -3)$. Take $R_1 = (1, -2, -3)$ and $R_2 = (-1, -2, -5)$.

$x = a_2b_3 - a_3b_2 = (-2)(-5) - (-3)(-2) = 10 - 6 = 4$

$y = a_3b_1 - a_1b_3 = (-3)(-1) - (1)(-5) = 3 + 5 = 8$

$z = a_1b_2 - a_2b_1 = (1)(-2) - (-2)(-1) = -2 - 2 = -4$

So $(x, y, z) = (4, 8, -4)$. Divide by 4: $\mathbf{X_1 = [\ 1, 2, -1\ ]^T}$.

Check: $AX_1 = [\ -2 + 4 + 3, 2 + 2 + 6, -1 - 4 + 0\ ]^T = [\ 5, 10, -5\ ]^T = 5X_1$. Correct.

For λ = −3. $A + 3I$ is:

$$A + 3I = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix}$$

Row 2 = 2 × Row 1 and Row 3 = −1 × Row 1. All rows are multiples, so we have only **one** equation: $x + 2y - 3z = 0$. (Cross multiplication would give (0, 0, 0). So we use Shortcut C.)

Here $a = 1$, $b = 2$, $c = -3$. Rule: $X = (b, -a, 0) = (2, -1, 0)$ and $X' = (c, 0, -a) = (-3, 0, -1)$.

Multiply X' by -1 to remove the minus signs: $\mathbf{X_2 = [\ 2, -1, 0\ ]^T}$ and $\mathbf{X_3 = [\ 3, 0, 1\ ]^T}$.

Check X_2: $2 + 2(-1) - 3(0) = 0$. Check X_3: $3 + 0 - 3(1) = 0$. Both satisfy the equation.

X_2 and X_3 are independent: X_2 has third entry 0 and X_3 has second entry 0, so neither is a multiple of the other.

Now X_1, X_2, X_3 are **three linearly independent** eigenvectors, so A is diagonalisable even though the eigenvalue -3 is repeated.

Modal matrix:

$$P = [\ X_1\ \ X_2\ \ X_3\ ] = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Step 3: Find P^{-1}

$|P| = 1(-1\cdot 1 - 0\cdot 0) - 2(2\cdot 1 - 0\cdot(-1)) + 3(2\cdot 0 - (-1)(-1)) = -1 - 4 - 3 = \mathbf{-8}$. (Not zero, so P^{-1} exists.)

The adjoint of P (transpose of the matrix of cofactors) is:

$$\text{adj } P = \begin{bmatrix} -1 & -2 & 3 \\ -2 & 4 & 6 \\ -1 & -2 & -5 \end{bmatrix}$$

Therefore $P^{-1} = \text{adj } P / |P| = (-1/8) \times \text{adj } P$:

$$P^{-1} = (-1/8)\begin{bmatrix} -1 & -2 & 3 \\ -2 & 4 & 6 \\ -1 & -2 & -5 \end{bmatrix}$$

Step 4: Find $D = P^{-1}AP$

Do it in two small parts. **Part (a): find AP.** We know $AX_1 = 5X_1$, $AX_2 = -3X_2$, $AX_3 = -3X_3$, so the columns of AP are $5X_1$, $-3X_2$, $-3X_3$. We can also check by multiplying:

$$AP = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -6 & -9 \\ 10 & 3 & 0 \\ -5 & 0 & -3 \end{bmatrix}$$

(Check one entry: first row, second column $= (-2)(2) + 2(-1) + (-3)(0) = -6$.)

Part (b): multiply by P^{-1}.

$$P^{-1}AP = (-1/8)\begin{bmatrix} -1 & -2 & 3 \\ -2 & 4 & 6 \\ -1 & -2 & -5 \end{bmatrix}\begin{bmatrix} 5 & -6 & -9 \\ 10 & 3 & 0 \\ -5 & 0 & -3 \end{bmatrix}$$

$$= (-1/8)\begin{bmatrix} -40 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 24 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} = \text{diag}(5, -3, -3)$$

Answer

$P^{-1}AP = \text{diag}(5, -3, -3)$. This is a diagonal matrix, and its diagonal numbers are the eigenvalues in the same order as the columns of P. **Verified.**

10. Example 3: find P, D and A^4

Question

(a) Find a matrix P which transforms A below to diagonal form. Hence calculate A^4. Also find the eigenvalues and eigenvectors of A.

(b) Determine the eigenvalues of A^{-1}.

(c) Diagonalise B below and hence find B^8.

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Part (a), Step 1: Find the eigenvalues of A (Shortcut A)

S_1 = 1 + 2 + 3 = 6.

S_2: cross out row 1, column 1: (2)(3) − (1)(2) = 4. Cross out row 2, column 2: (1)(3) − (−1)(2) = 5. Cross out row 3, column 3: (1)(2) − (0)(1) = 2. So S_2 = 4 + 5 + 2 = 11.

S_3 = |A| = 1(6 − 2) − 0 + (−1)(2 − 4) = 4 + 2 = 6.

The equation is **$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$**.

Trial: try λ = 1: 1 − 6 + 11 − 6 = 0. So λ = 1 is a root. Divide:

Coefficients	1	−6	11	−6
Multiply by 1		1	−5	6
Add	1	−5	6	0

The quadratic is $\lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3)$. **Eigenvalues of A: λ = 1, 2, 3.** All are different, so we will get three independent eigenvectors.

Check: sum = 6 = S_1. Product = 6 = S_3.

Part (a), Step 2: Find the eigenvectors (Shortcut B)

For λ = 1. A − I is:

$$A - I = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix}$$

Rows 2 and 3 are multiples, so choose rows 1 and 2: R_1 = (0, 0, −1) and R_2 = (1, 1, 1).

$x = a_2b_3 - a_3b_2 = (0)(1) - (-1)(1) = 1$

$y = a_3b_1 - a_1b_3 = (-1)(1) - (0)(1) = -1$

$z = a_1b_2 - a_2b_1 = (0)(1) - (0)(1) = 0$

X_1 = [1, −1, 0]^T.

For λ = 2. $A - 2I$ is:

$$A - 2I = \begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix}$$

Rows 1 and 2 are multiples, so choose rows 1 and 3: $R_1 = (-1, 0, -1)$ and $R_2 = (2, 2, 1)$.

$x = a_2b_3 - a_3b_2 = (0)(1) - (-1)(2) = 2$

$y = a_3b_1 - a_1b_3 = (-1)(2) - (-1)(1) = -2 + 1 = -1$

$z = a_1b_2 - a_2b_1 = (-1)(2) - (0)(2) = -2$

So $(x, y, z) = (2, -1, -2)$. Multiply by -1: $\mathbf{X_2 = [\ -2, 1, 2\ ]^T}$.

For λ = 3. $A - 3I$ is:

$$A - 3I = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$$

Choose rows 1 and 2: $R_1 = (-2, 0, -1)$ and $R_2 = (1, -1, 1)$.

$x = a_2b_3 - a_3b_2 = (0)(1) - (-1)(-1) = -1$

$y = a_3b_1 - a_1b_3 = (-1)(1) - (-2)(1) = -1 + 2 = 1$

$z = a_1b_2 - a_2b_1 = (-2)(-1) - (0)(1) = 2$

$\mathbf{X_3 = [\ -1, 1, 2\ ]^T}$.

Quick check of X_2: $AX_2 = [\ 1(-2) + 0 + (-1)(2), 1(-2) + 2(1) + 1(2), 2(-2) + 2(1) + 3(2)\ ]^T = [\ -4, 2, 4\ ]^T = 2X_2$. Correct.

Part (a), Step 3: Modal matrix P and P^{-1}

$$P = [\,X_1 \;\; X_2 \;\; X_3\,] = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

$|P| = 1(1\cdot 2 - 1\cdot 2) - (-2)((-1)\cdot 2 - 1\cdot 0) + (-1)((-1)\cdot 2 - 1\cdot 0) = 0 - 4 + 2 = \mathbf{-2}$. Not zero, so P^{-1} exists.

$$\text{adj } P = \begin{bmatrix} 0 & 2 & -1 \\ 2 & 2 & 0 \\ -2 & -2 & -1 \end{bmatrix} \qquad \text{so} \qquad P^{-1} = (-1/2)\begin{bmatrix} 0 & 2 & -1 \\ 2 & 2 & 0 \\ -2 & -2 & -1 \end{bmatrix}$$

Part (a), Step 4: Check that $P^{-1}AP$ is diagonal

As before, the columns of AP are $1\cdot X_1$, $2\cdot X_2$, $3\cdot X_3$:

$$AP = \begin{bmatrix} 1 & -4 & -3 \\ -1 & 2 & 3 \\ 0 & 4 & 6 \end{bmatrix}$$

Multiply by $P^{-1} = (-1/2)\,\text{adj } P$:

$$P^{-1}AP = (-1/2)\begin{bmatrix} 0 & 2 & -1 \\ 2 & 2 & 0 \\ -2 & -2 & -1 \end{bmatrix}\begin{bmatrix} 1 & -4 & -3 \\ -1 & 2 & 3 \\ 0 & 4 & 6 \end{bmatrix} = (-1/2)\begin{bmatrix} -2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

So $\mathbf{D = diag(1, 2, 3)}$. A is diagonalised.

Part (a), Step 5: Find A^4

$D^4 = \text{diag}(1^4, 2^4, 3^4) = \text{diag}(1, 16, 81)$.

$A^4 = P\,D^4\,P^{-1}$. First $P\,D^4$ (multiply column 1 by 1, column 2 by 16, column 3 by 81):

$$P\,D^4 = \begin{bmatrix} 1 & -32 & -81 \\ -1 & 16 & 81 \\ 0 & 32 & 162 \end{bmatrix}$$

Multiply by $P^{-1} = (-1/2)\,\text{adj } P$:

$$A^4 = (-1/2)\begin{bmatrix} 1 & -32 & -81 \\ -1 & 16 & 81 \\ 0 & 32 & 162 \end{bmatrix}\begin{bmatrix} 0 & 2 & -1 \\ 2 & 2 & 0 \\ -2 & -2 & -1 \end{bmatrix} = (-1/2)\begin{bmatrix} 98 & 100 & 80 \\ -130 & -132 & -80 \\ -260 & -260 & -162 \end{bmatrix}$$

(For example, first row, first column: $1(0) + (-32)(2) + (-81)(-2) = -64 + 162 = 98$.) Dividing every number by -2:

Answer for (a)

$$A^4 = \begin{bmatrix} -49 & -50 & -40 \\ 65 & 66 & 40 \\ 130 & 130 & 81 \end{bmatrix}$$

Part (b): Eigenvalues of A^{-1}

The eigenvalues of A are 1, 2, 3. None is zero, and $|A| = 6 \neq 0$, so A^{-1} exists. Start from $AX = \lambda X$ and multiply by A^{-1} on the left:

$$X = \lambda A^{-1}X \quad \Rightarrow \quad A^{-1}X = (1/\lambda)\, X$$

So X is also an eigenvector of A^{-1}, and the eigenvalue is **$1/\lambda$**. Hence the eigenvalues of A^{-1} are:

Answer for (b)

1, 1/2, 1/3 (with the same eigenvectors X_1, X_2, X_3).

Part (c): Diagonalise B and find B^8

$$B = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Eigenvalues (Shortcut A).

S_1 = 1 + 2 + 3 = 6.

S_2: cross out row 1, column 1: (2)(3) − (0)(0) = 6. Cross out row 2, column 2: (1)(3) − (1)(0) = 3. Cross out row 3, column 3: (1)(2) − (6)(1) = −4. So S_2 = 6 + 3 − 4 = 5.

S_3 = |B| = 3[(1)(2) − (6)(1)] = 3(−4) = −12 (expand along the last row).

The equation is **$\lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0$**. Trial: λ = −1: −1 − 6 − 5 + 12 = 0. So λ = −1 is a root. Divide:

Coefficients	1	−6	5	12
Multiply by −1		−1	7	−12
Add	1	−7	12	0

The quadratic is $\lambda^2 - 7\lambda + 12 = (\lambda - 3)(\lambda - 4)$. **Eigenvalues of B: λ = −1, 3, 4.** (Three different values, so B is diagonalisable.)

Check: sum = −1 + 3 + 4 = 6 = S_1. Product = (−1)(3)(4) = −12 = S_3.

Eigenvectors (Shortcut B).

For λ = −1. B + I = [[2, 6, 1], [1, 3, 0], [0, 0, 4]]. Choose R_1 = (1, 3, 0) and R_2 = (0, 0, 1) (row 3 divided by 4).

$x = a_2b_3 - a_3b_2 = (3)(1) - (0)(0) = 3.$
$y = a_3b_1 - a_1b_3 = (0)(0) - (1)(1) = -1.$
$z = a_1b_2 - a_2b_1 = (1)(0) - (3)(0) = 0.$

(3, −1, 0) multiplied by −1: **X_1 = [−3, 1, 0]^T.**

For λ = 3. B − 3I = [[−2, 6, 1], [1, −1, 0], [0, 0, 0]]. Choose R_1 = (−2, 6, 1) and R_2 = (1, −1, 0).

$x = a_2b_3 - a_3b_2 = (6)(0) - (1)(-1) = 1.$
$y = a_3b_1 - a_1b_3 = (1)(1) - (-2)(0) = 1.$
$z = a_1b_2 - a_2b_1 = (-2)(-1) - (6)(1) = -4.$

X_2 = [1, 1, −4]^T.

For λ = 4. $B - 4I = [[-3, 6, 1], [1, -2, 0], [0, 0, -1]]$. Choose $R_1 = (1, -2, 0)$ and $R_2 = (0, 0, -1)$.

$x = a_2b_3 - a_3b_2 = (-2)(-1) - (0)(0) = 2.$

$y = a_3b_1 - a_1b_3 = (0)(0) - (1)(-1) = 1.$

$z = a_1b_2 - a_2b_1 = (1)(0) - (-2)(0) = 0.$

$X_3 = [\ 2, 1, 0\ ]^T$.

Quick check for X_2: $BX_2 = [\ 1 + 6 - 4, 1 + 2, -12\ ]^T = [\ 3, 3, -12\ ]^T = 3X_2$. Correct.

$$P = \begin{bmatrix} -3 & 1 & 2 \\ 1 & 1 & 1 \\ 0 & -4 & 0 \end{bmatrix} \qquad D = \operatorname{diag}(-1, 3, 4)$$

Find P^{-1}. $|P| = -3(0+4) - 1(0-0) + 2(-4-0) = -12 - 8 = \mathbf{-20}$.

$$\text{adj } P = \begin{bmatrix} 4 & -8 & -1 \\ 0 & 0 & 5 \\ -4 & -12 & -4 \end{bmatrix} \qquad P^{-1} = (-1/20)\begin{bmatrix} 4 & -8 & -1 \\ 0 & 0 & 5 \\ -4 & -12 & -4 \end{bmatrix}$$

Find $B^8 = P\,D^8\,P^{-1}$. $D^8 = \text{diag}((-1)^8, 3^8, 4^8) = \text{diag}(1, 6561, 65536)$. Multiply the columns of P by 1, 6561, 65536:

$$P\,D^8 = \begin{bmatrix} -3 & 6561 & 131072 \\ 1 & 6561 & 65536 \\ 0 & -26244 & 0 \end{bmatrix}$$

Now multiply by $P^{-1} = (-1/20)\,\text{adj } P$:

$$B^8 = (-1/20)\begin{bmatrix} -3 & 6561 & 131072 \\ 1 & 6561 & 65536 \\ 0 & -26244 & 0 \end{bmatrix}\begin{bmatrix} 4 & -8 & -1 \\ 0 & 0 & 5 \\ -4 & -12 & -4 \end{bmatrix}$$

$$= (-1/20)\begin{bmatrix} -524300 & -1572840 & -491480 \\ -262140 & -786440 & -229340 \\ 0 & 0 & -131220 \end{bmatrix}$$

Answer for (c)

$$B^8 = \begin{bmatrix} 26215 & 78642 & 24574 \\ 13107 & 39322 & 11467 \\ 0 & 0 & 6561 \end{bmatrix}$$

Quick sense check: the last row of B is (0, 0, 3), so the last row of B^8 must be $(0, 0, 3^8) = (0, 0, 6561)$. It is.

11. Summary, common mistakes and practice

Summary: lines to remember

1. A is diagonalisable if $P^{-1}AP = D$ (diagonal) for some invertible P.

2. P = modal matrix (columns = eigenvectors). D = spectral matrix (diagonal = eigenvalues, same order).

3. A ($n \times n$) is diagonalisable exactly when it has n linearly independent eigenvectors.

4. n distinct eigenvalues ⇒ always diagonalisable. Repeated eigenvalues ⇒ must check.

5. Real symmetric matrix with distinct eigenvalues: eigenvectors are orthogonal, and $P^{-1} = P^{T}$ if we use unit vectors.

6. $A^{n} = P\,D^{n}\,P^{-1}$, with $D^{n} = \mathrm{diag}(\lambda_1^{n}, \ldots, \lambda_n^{n})$.

7. Eigenvalues (3×3): solve $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ by one trial root and the division table.

8. Eigenvectors: cross-multiply two rows of $(A - \lambda I)$. If only one equation is left, use $(b, -a, 0)$ and $(c, 0, -a)$.

Common mistakes

- Writing $P^{-1}AP$ as $AP^{-1}P$ or $P\,A\,P^{-1}$. The **order** is P^{-1} on the left and P on the right.
- Putting the eigenvalues in D in a different order from the eigenvectors in P.
- Taking $X = 0$ as an eigenvector. An eigenvector must be non-zero.
- Forgetting to check that the eigenvectors are independent when an eigenvalue is repeated.
- Using $A^{n} = P^{-1}\,D^{n}\,P$. This is wrong. The correct formula is $A^{n} = P\,D^{n}\,P^{-1}$.

Practice questions

Q1. Diagonalise the matrix below and find A^5.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

Q2. Is the matrix below diagonalisable? Give a reason.

$$B = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Answers

Q1. Eigenvalues 3 and −1. Eigenvectors $X_1 = [1, 1]^T$ for $\lambda = 3$ and $X_2 = [1, -1]^T$ for $\lambda = -1$. $P = [[1, 1], [1, -1]]$, $D = \mathrm{diag}(3, -1)$. Here A is symmetric, so we can also use unit vectors and $P^{-1} = P^{T}$. $A^5 = P\,D^5\,P^{-1}$ with $D^5 = \mathrm{diag}(243, -1)$, which gives $A^5 = [[121, 122], [122, 121]]$.

Q2. Eigenvalue $\lambda = 2$ (repeated twice). $(B - 2I)X = 0$ gives $y = 0$, so the only eigenvectors are multiples of $[1, 0]^T$. We get only **one** independent eigenvector, but we need two. So B is **not** diagonalisable.